DAY — 08 SEAT NUMBER

2011	III	07	1100	J - 554	(E)

PHYSICS PAPER - I (54)

(Mechanics, Properties of Matter, Sound and Heat)

Time : 2 Hrs. (4 Pages) Max. Marks : 40

Note : (i) All questions are compulsory.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of only logarithmic table is allowed.

(v) All symbols have their usual meaning unless otherwise stated.

(vi) Answer to every question must be written on a new page.

Q. 1. Select and write the most appropriate answer from the given alternatives for each sub-question : [8]

(i) A sonometer wire is vibrating in second overtone. The number of nodes and antinodes formed respectively are ..... . (1)

(a) 2, 3 (b) 3, 4

(c) 4, 3 (d) 3, 2

(ii) The angle between particle velocity and wave velocity in transverse wave is ..... . (1)

(a) Zero rad (b) $\frac{\pi}{2}$ rad

(c) $\frac{\pi}{4}$ rad (d) π rad

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(iii) In equilibrium the tensile stress to which a wire of radius 'r' is subjected by attaching a mass 'm' is ..... . (1)

(a) $\frac{mg}{\pi r}$ (b) $\frac{mg}{2\pi r}$

(c) $\frac{mg}{\pi r^2}$ (d) $\frac{mg}{2\pi r^2}$

(iv) The displacement of a particle performing linear S. H. M. in one time period is ..... . (1)

(a) a (b) 2a

(c) 0 (zero) (d) 4a

(v) The radius of gyration of a ring of radius 'R' about a tangent perpendicular to its plane is ..... . (1)

(a) $\sqrt{2}R$ (b) 2R

(c) $\frac{R}{\sqrt{2}}$ (d) $\frac{R}{2}$

(vi) As we go from the equator towards the poles of the Earth, the value of acceleration due to gravity ..... . (1)

(a) remains constant (b) decreases

(c) increases (d) decreases upto latitude of 45°

(vii) If a cycle wheel of radius 0·4 m completes one revolution in 2 seconds, then acceleration of the cycle is ..... . (1)

(a) $0{\cdot}4\pi$ m/s^2 (b) $0{\cdot}4\pi^2$ m/s^2

(c) $\frac{\pi^2}{0{\cdot}4}$ m/s^2 (d) $\frac{0{\cdot}4}{\pi^2}$ m/s^2

(viii) The absolute temperature of a gas is increased three times. The root mean square velocity of the gas molecules will be ..... . (1)

(a) 3 times (b) 9 times

(c) $\frac{1}{3}$ times (d) $\sqrt{3}$ times

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Q. 2. (A) Attempt any ONE : [8]

(i) A steel wire is acted upon by a load of 10 N. Calculate the extension produced in the wire if the strain energy stored in the wire is $1{\cdot}1 \times 10^{-3}$ J. (2)

(ii) Calculate the kinetic energy of helium molecules in 1 cm^3 at pressure 2 atmosphere. (1 atmosphere = $1{\cdot}013 \times 10^5$ N/m^2) (2)

(B) Attempt any TWO :

(i) Show that total energy of a body performing vertical circular motion is conserved. (3)

(ii) Show that for a small amplitude, the motion of a simple pendulum is linear S. H. M. Hence find its period. (3)

(iii) Derive an expression for the moment of inertia of a thin uniform disc about an axis passing through its centre and perpendicular to its plane. (3)

Q. 3. (A) Attempt any ONE : [8]

(i) What is capillarity? Hence state its any 'two' applications. (2)

(ii) State Wien's displacement law for black body and define solar constant. (2)

(B) Attempt any TWO :

(i) State any 'four' assumptions of kinetic theory of gases and define R. M. S. velocity. (3)

(ii) Show that linear S. H. M. is the projection of uniform circular motion on any diameter. (3)

(iii) Describe Searle's method to determine Young's modulus of the material of wire. (3)

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Q. 4. (A) Attempt any TWO : [8]

(i) State and prove the law of conservation of angular momentum. (2)

(ii) Define centripetal force and give its any 'two' examples. (2)

(iii) Draw the neat labelled diagrams of Melde's experiment for the same set up showing the number of loops in
(a) Parallel position
(b) Perpendicular position (2)

(B) Attempt any ONE :

(i) Define binding energy and hence derive its expression for a satellite revolving at certain height above the Earth's surface. (4)

(ii) Explain the formation of concave and convex surface of liquid on the basis of molecular forces. (4)

Q. 5. Attempt any TWO : [8]

(i) A copper ball cools from 62°C to 50°C in 10 minutes and to 42°C in the next 10 minutes. Calculate the temperature of the surroundings. What is the temperature of the ball at the end of next 8 minutes. (4)

(ii) In a resonance tube experiment the shortest length of air column is resonating with a tuning fork of frequency 300 Hz is 25 cm. The next resonance occurs when water level is 80 cm below the open end. Find the velocity of sound in air and diameter of the tube. (4)

(iii) A set of 12 tuning forks is arranged in the increasing order of frequencies. Each tuning fork produces 'y' beats per second with previous one. The last tuning fork is an octave of the first. The fifth tuning fork has frequency 90 Hz. Find 'y'. Hence find the frequency of the last fork. (4)

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